

ASSIGNMENT 3

DBMS Fundamentals – Local Case Study

Name: Rima Maji

Roll No.: 150096725215

Cohort: Demis Hassabis

Reflection

1. Current Method Summary

The selected business is a small local kirana shop that sells groceries and daily-use household products. The shop maintains records related to daily sales, stock, customer dues and supplier purchases. At present, most of the information is maintained manually using notebooks, registers and bills. Daily sales are recorded in a notebook, while bills are kept separately for reference. Stock is checked manually by looking at the products available in the shop. Customer credit and outstanding dues are also recorded manually. Supplier purchase details are maintained using bills and written records. When an old transaction is required, the owner has to search through the notebooks and bills.

2. Flaws & Limitations

The manual method has several limitations. Maintaining a large number of records in notebooks can result in missed, duplicate or incorrect entries. Searching for an old sale or customer payment takes time because the owner has to manually search through previous records. Stock quantities may also become inaccurate because stock is checked manually.

Customer dues have to be updated manually, which can result in errors or missed payments.

Another major problem is the lack of proper backup. If a notebook or bill is lost or damaged, it can become difficult to recover the information. The records also have limited security because physical notebooks can potentially be accessed or changed by anyone who has access to them. These problems show the need for a more organized database system.

3. Proposed DBMS Solution & Component Mapping

A relational DBMS can organize the shop's information into separate tables such as **Products, Sales, Customers/Dues and Suppliers**. This would reduce unnecessary duplication and make information easier to search and update.

The **Query Processor** would allow the owner to quickly search for old sales, customer dues and available stock. The **DDL/DML Interpreter** would handle operations such as creating database structures and inserting, updating or deleting records. The **Transaction Manager** would ensure that related operations are completed correctly; for example, when a product is sold, the sale can be recorded and the stock quantity updated consistently.

The **Storage Manager** would manage the stored business records in the database. The **Buffer Manager** would temporarily keep frequently accessed data in memory while database operations are being performed. The **Authorization/Security Subsystem** would control which users are allowed to access or modify particular records.

Therefore, using a DBMS would make the shop's record management more organized, easier to search, more secure and less dependent on physical notebooks and bills.

ANNEXURE – INTERVIEW NOTES

Business Type: Local Kirana / General Store

Name - Rima Mafi

Roll No - 150096725215

Co-host - Denis Hassabir

Type of Business - General store

Interview

1) How do you currently manage your shop sales?
→ Daily sales are recorded in a notebook and some bills are kept separately for reference.

2) How do you keep track of your stock?
→ The products on the shelves are checked manually and items that are running low are noted down.

3) How do you manage customers who buy on credit?
→ The customer's name, amount due and payment details are recorded in a notebook and updated when the customer makes a payment.

4) How do you ~~store your~~ record purchases from suppliers?
→ Supplier bills are kept and important purchase details such as the product, quantity and price are recorded.

5) Where do you store your records?
→ Most of the information is stored in notebooks and some bills are kept separately.

6) What problems do you face while maintaining these records?
→ Sometimes entries can be missed or written incorrectly. It also takes time to find old sales or customer payment records.

7) What happens if a notebook or bill is lost or damaged?
→ It becomes difficult to recover the information because there is no proper backup.

8) How do you find an old ~~transact~~ transaction?
→ The notebook and old bills have to be searched manually.

9) Would a computerized database system be useful for the shop?
→ Yes, it would make it easier to search records and manage the information.

DBMS Component Diagram

